Practice · scanf

Each question lists its options; the right one is marked **Right answer**. *Feedback* is what students see after they answer. In a typed question the blank is in square brackets, with the accepted answers separated by `|`.

(mc) L1 · What scanf does

What does `scanf` do?

- It prints a value on the screen.
- **Right answer:** It waits for you to type something and stores it in a variable.
- It makes a new variable.
- It compiles the program.

Feedback: `scanf` reads from the keyboard. The program stops and waits until you type and press Enter, then puts what you typed into the variable.

(mc) L1 · The & sign

In `scanf("%d", &age);`, why is there an `&` before `age`?

- It is a typo.
- It means "and".
- **Right answer:** It tells `scanf` where the variable `age` is, so it can put the value in it.
- It makes `age` a `double`.

Feedback: `scanf` has to put a value into the box, so it needs to know where the box is. `&age` means "the place where age is". Without `&` the program compiles but runs wrong.

(mc) L1 · Reading a double

Which hole reads a `double` with `scanf`?

- `%f`
- `%d`
- **Right answer:** `%lf`
- `%.2f`

Feedback: For `scanf`, a `double` needs `%lf` (with the letter L). `%f` and `%.2f` are for `printf`. Yes, this is annoying. It is just how C is.

(mc) L2 · What is stored

The program runs this:

```
int age;
double height;
scanf("%d", &age);
scanf("%lf", &height);
```

You type `19`, press Enter, type `1.82`, press Enter. What is stored?

- **Right answer:** `age` holds 19 and `height` holds 1.82.
- `age` holds 1.82 and `height` holds 19.

- Both hold 19.
- Nothing, because there are two `scanf` lines.

Feedback: The first `scanf` takes the first number you typed and puts it in `age`. The second takes the next number and puts it in `height`.

(mc) L2 · The missing &

What happens when you compile and run this?

```
int n;
scanf("%d", n);
printf("%d\n", n);
```

- It works normally.
- The compiler refuses to make a program.
- **Right answer:** It compiles, and with `-Wall` there is a warning; when it runs it crashes or prints a nonsense number.
- It prints `n`.

Feedback: Without `&`, `scanf` does not know where `n` is. The compiler still makes a program, but only warns you if you used `-Wall`. Running it gives a crash or garbage.

(mc) L2 · Why no newline in the question

Why is there no `\n` in `printf("Enter age: ");` before a `scanf`?

- Because `\n` is not allowed before `scanf`.
- **Right answer:** So the cursor stays on the same line and you type right after the colon.
- Because `scanf` adds the `\n`.
- It is a mistake; `\n` is required.

Feedback: Without `\n` the cursor waits on the same line, which looks like a proper question. With `\n` you would type on the next line. Both work; this one looks better.

(fib) L3 · Read a whole number

Complete the line that reads a whole number into the `int` variable `age`: scanf("%d", [&age]);

Feedback: `&age`. The `&` gives `scanf` the place where `age` is.

(fib) L3 · Read a decimal number

Complete the line that reads a number with a decimal point into the `double` variable `price`: scanf("[%lf]", &price);

Feedback: `%lf`. For `scanf`, a `double` always needs `%lf`.

(fib) L3 · Read two numbers

Complete the line that reads two whole numbers into `a` and `b`: scanf("%d %d", [&a, &b|&a,&b]);

Feedback: `&a, &b`. Two holes, two variables, each with its own `&`.